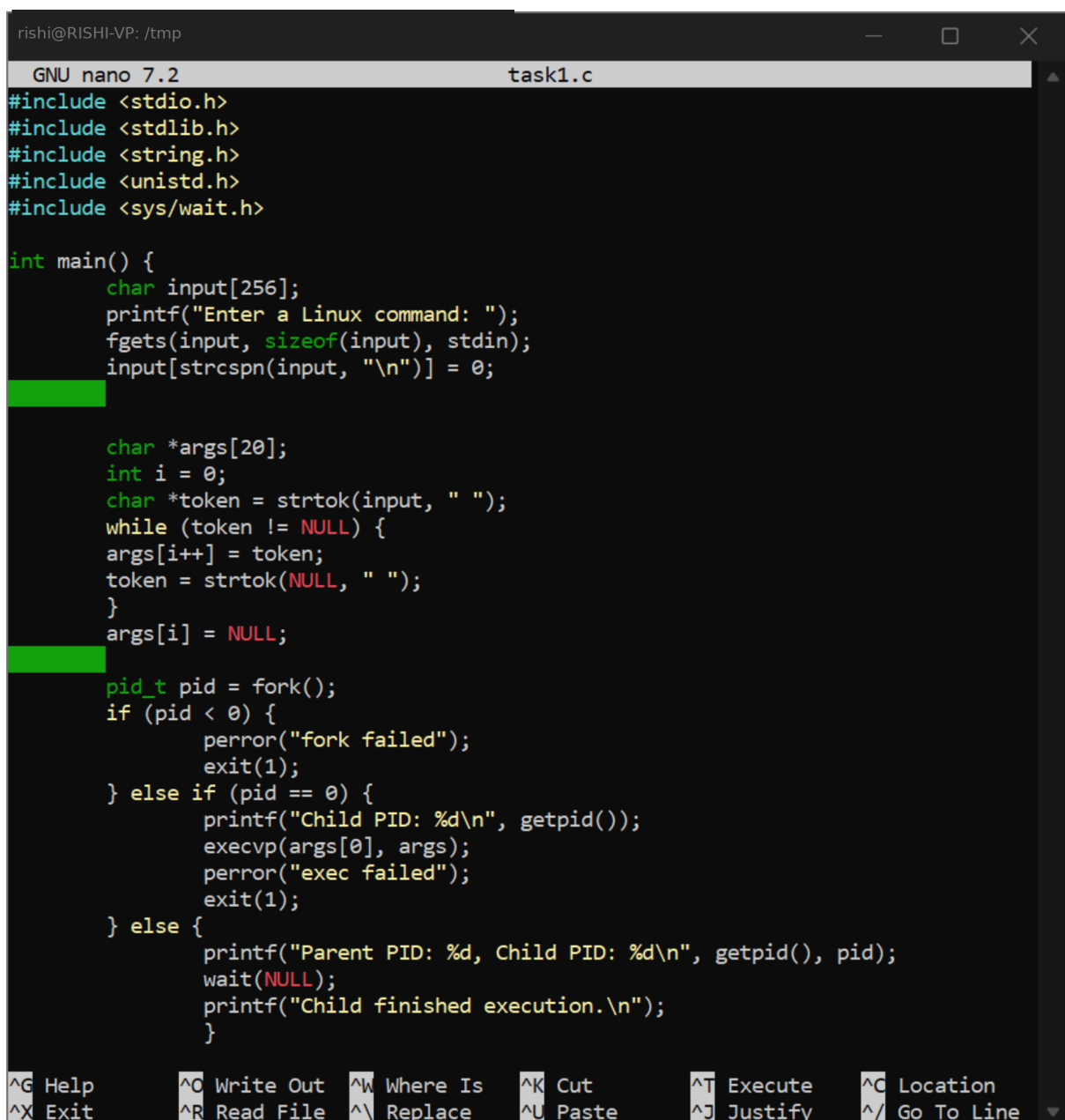


TASK 1

1. Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that 1. Accept a Linux command as input. 2. Create a child process using fork(). 3. Execute the command in the child process using an appropriate exec() system call. 4. Allow the parent process to wait for the child using wait (). 5. Display the Process ID (PID) of both parent and child processes.

Using Linux terminal commands (uname, lscpu, lsblk, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.



```
rishi@RISHI-VP: /tmp
GNU nano 7.2 task1.c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {
    char input[256];
    printf("Enter a Linux command: ");
    fgets(input, sizeof(input), stdin);
    input[strcspn(input, "\n")] = 0;

    char *args[20];
    int i = 0;
    char *token = strtok(input, " ");
    while (token != NULL) {
        args[i++] = token;
        token = strtok(NULL, " ");
    }
    args[i] = NULL;

    pid_t pid = fork();
    if (pid < 0) {
        perror("fork failed");
        exit(1);
    } else if (pid == 0) {
        printf("Child PID: %d\n", getpid());
        execvp(args[0], args);
        perror("exec failed");
        exit(1);
    } else {
        printf("Parent PID: %d, Child PID: %d\n", getpid(), pid);
        wait(NULL);
        printf("Child finished execution.\n");
    }
}
```

^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^E Replace ^U Paste ^J Justify ^_ Go To Line

OUTPUT :

```
rishi@RISHI-VP: /tmp

root@PC:~# su sairam
rishi@RISHI-VP:/root$ nano task1.c
rishi@RISHI-VP:/root$ pwd
/root
rishi@RISHI-VP:/root$ whoami
sairam
rishi@RISHI-VP:/root$ ls -ld
ls: cannot access '.': Permission denied
rishi@RISHI-VP:/root$ cd /tmp
rishi@RISHI-VP:/tmp$ nano task1.c
rishi@RISHI-VP:/tmp$ gcc cmd_exec.c -o cmd_exec
cc1: fatal error: cmd_exec.c: No such file or directory
compilation terminated.
rishi@RISHI-VP:/tmp$ gcc task1.c -o task1
rishi@RISHI-VP:/tmp$ ./task1

Enter a Linux command: ls -l
Parent PID: 979, Child PID: 982
Child PID: 982
snap-private-tmp
systemd-private-4bc431fa2fe54f6ebd6553fa9f0b9631-systemd-logind.service-eKlpVF
systemd-private-4bc431fa2fe54f6ebd6553fa9f0b9631-systemd-resolved.service-slbyfB
systemd-private-4bc431fa2fe54f6ebd6553fa9f0b9631-systemd-timesyncd.service-MZohCu
task1
task1.c
Child finished execution.
rishi@RISHI-VP:/tmp$ pwd
/tmp
rishi@RISHI-VP:/tmp$ date
Wed Aug  5 03:34:48 UTC 2026
rishi@RISHI-VP:/tmp$ nano task1
rishi@RISHI-VP:/tmp$ nano task1.c
```